Banapuram Pranay Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.93
BHAVAN'S SRI AUROBINDO JUNIOR COLLEGE	Class XII (TGBIE)	2024	98.80%
ATOMIC ENERGY CENTRAL SCHOOL - 2 , HYDERABAD	Class X (CBSE)	2022	95.20%

SKILLS

Programming languages : Python , C , MATLAB , Verilog , LATEX

Libraries and Frameworks : NumPy , Pandas , MatPlotlib ,PyTorch, OpenCV ,Scikit-Learn,Scipy,Librosa

Machine Learning : Supervised Learning , Deep Learning , Model Training ad Evaluation

Computer Vision : Image processing , audio processing , Feature Extraction , Audio-Visual Learning

Tools and Platforms : Git , Github , Linux(Ubuntu) ,Microsoft Excel, Google Colab

Embedded system and Circuit Design : Arduino Uno , LTspice , Quartus Prime , DE10-SoC FPGA board

RELEVANT COURSES

Mathematics : Math for EE(Linear Algebra,Calculus) , Differential Equations and Transform Techniques, Probability,Random Processes,Matrix Theory

Computer Science and AI : Introduction to Programming(C) , Data Structures and Algorithms , Pattern Recognition and Machine Learning,Scientific Programming for EE

Electrical Labs : Electric Circuits Lab , EDC Lab , DSP Lab,Digital SystemsLab

Electrical and Electronics : Signal Processing , Digital Systems , Analog Electronics , Control Systems,Power systems ,Electromagnetics , Energy Management

PROJECTS

AVZoom: Audio-Visual Speaker Enhancement

Technologies: Python, PyTorch, OpenCV, NumPy

- Developed an audio-visual deep learning system to enhance the speech of a target speaker using synchronized video and audio inputs.
- Built data preprocessing pipelines for extracting and aligning visual and audio features.
- Implemented and trained convolutional neural network models for multimodal speech enhancement.
- Evaluated model performance using objective speech quality metrics.

Real-Time 3D Pose Estimation using ArUco Markers

Technologies:Python, OpenCV, NumPy

- Developed a real-time computer vision pipeline to detect ArUco markers from a monocular camera and estimate their 3D pose using PnP-based pose estimation.

- Implemented 3D-to-2D projection to overlay a virtual cube on the detected marker and visualized its **X, Y, Z position** in real time.

- Worked with camera intrinsic parameters, coordinate transformations, marker geometry, and OpenCV-based 3D vision.

Synthetic Speech Attribution

Technologies: Python, PyTorch, NumPy

- Developed a deep learning model to identify the source of synthetic speech.

- Extracted and analyzed acoustic features from speech recordings.

- Trained and evaluated neural network models for speech classification.

- Investigated model performance across different synthetic speech generators.

Eigenvalue Computation using Hybrid Newton-Power Method

Technologies: C

- Designed and implemented a hybrid numerical algorithm combining the Power Method and Newton's Method for eigenvalue computation.

- Developed the complete implementation in C without using external numerical or linear algebra libraries.

- Implemented matrix operations, convergence criteria, and iterative optimization from first principles.

- Evaluated numerical accuracy and convergence characteristics on multiple test matrices.

8-Point Radix-2 FFT Processor on FPGA

Technologies: Verilog HDL, MATLAB, Intel Quartus Prime

- Designed and implemented a pipelined streaming 8-point Radix-2 FFT processor in Verilog HDL.

- Developed a streaming architecture accepting one complex input sample per clock cycle.

- Verified functional correctness by comparing hardware outputs with MATLAB implementation and Intel FFT IP.

- Evaluated throughput, maximum operating frequency (Fmax), and hardware resource utilization.

- Optimized the pipelined architecture to achieve performance comparable to the vendor-provided FFT IP core.

EXTRACURRICULAR

VideoMaking : Participated in film production, videography, and creative media projects.

Public Speaking : QuarterFinalist in Yuva-Sangarsh (Inter Collegiate Leadership Competition)

Third Place, Gyan Setu Quiz - Indian Knowledge Systems (IKS) Student Chapter

POSITIONS OF RESPONSIBILITY

Core Member, Behind The Lens - Film Making Club